Euclidean Circle-and-Line Partitions with Finite Complement and an Elliptic Extension

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October 2026

Abstract

Let $F \subset \mathbb{R}^2$ be finite. We first classify partitions of $\mathbb{R}^2 \setminus F$ into pairwise disjoint complete Euclidean circles and affine lines. Such a partition exists if and only if $|F| \leq 2$. Every circular component is a once-punctured plane or half-plane, and its circle leaves admit a unique radius parametrization

$$C_r = \partial B(c(r), r), \qquad |c(s) - c(r)| < s - r \quad (0 < r < s),$$

with $c(r) \to p$ at the puncture. The increasing disk union $K_c = \bigcup_{r>0} B(c(r), r)$ is necessarily either the plane or a half-plane, yielding a complete four-type global classification. We then show a sharp contrast after ellipses are admitted: for every n-point set F, the complement $\mathbb{R}^2 \setminus F$ can be partitioned using only $n-1$ parallel lines and noncircular ellipses. Among partitions using ellipses and finitely many lines, $n-1$ line leaves are necessary. Thus the at-most-two-puncture obstruction is a specifically circular phenomenon and disappears once variable eccentricity is available.

1 Introduction

A *complete Euclidean circle* is a set

$$\partial B(c, r) = \{x \in \mathbb{R}^2 : |x - c| = r\}, \qquad r > 0,$$

and a *complete line* is an entire affine line. Given a finite set $F \subset \mathbb{R}^2$, our first problem is to classify partitions of $\mathbb{R}^2 \setminus F$ whose members are complete circles or complete lines. We then enlarge the allowed compact leaves from circles to arbitrary nondegenerate ellipses; since ellipses are nondegenerate real conics, this also settles the corresponding existence problem in which circles, ellipses, parabolas, hyperbolas, and lines are all allowed.

The circle-and-line problem combines elementary Euclidean geometry with the topology of circle decompositions of surfaces. Moussong and Simányi proved that a connected surface admitting a decomposition into Jordan circles belongs to one of seven topological types; in particular, an orientable boundaryless such surface is either a torus or an open annulus [2]. Their theorem applies to arbitrary circle decompositions, not merely to foliations, which is important here because no regular dependence of the leaves is assumed. A closely related circle-only question appeared on Mathematics Stack Exchange in 2016; the two-puncture case was discussed there, together with the Apollonius family as a near miss leaving one line uncovered [3].

Our main results are as follows.

Theorem 1.1 (Circle–line classification, summary)**.** *Let $F \subset \mathbb{R}^2$ be finite. Then $\mathbb{R}^2 \setminus F$ admits a partition into pairwise disjoint complete Euclidean circles and affine lines if and only if*

$$|F| \leq 2.$$

Moreover, up to Euclidean isometry, every such partition is one of the four types listed in Theorem 6.1: the all-lines partition when $F = \varnothing$; a pure circular partition when $|F| = 1$; a one-sided circular half-plane together with a parallel half-plane of line leaves when $|F| = 1$; or two circular half-planes separated by a possibly degenerate closed strip of parallel line leaves when $|F| = 2$.

The metric freedom inside each circular region is itself completely describable.

Theorem 1.2 (Admissible-center parametrization, summary)**.** *Fix $p \in \mathbb{R}^2$. Every Euclidean circle decomposition of a once-punctured plane or once-punctured open half-plane has a unique parametrization by Euclidean radius,*

$$C_r = \partial B(c(r), r), \qquad r > 0,$$

where

$$\lim_{r \downarrow 0} c(r) = p, \qquad |c(s) - c(r)| < s - r \quad (0 < r < s).$$

Conversely, every map with these two properties partitions

$$K_c \setminus \{p\}, \qquad K_c := \bigcup_{r>0} B(c(r), r),$$

into the circles C_r. The domain K_c is necessarily either $\mathbb{R}^2$ or an open half-plane. If

$$d(r) := r - |c(r) - p|,$$

then $K_c = \mathbb{R}^2$ exactly when $d(r) \to \infty$; otherwise $d(r) \to \delta < \infty$, K_c is a half-plane, and $\delta = \operatorname{dist}(p, \partial K_c)$.

Thus the circle-and-line problem is topologically rigid but metrically infinite-dimensional: the center curve need not be concentric or coaxal. The strict nesting inequality also has a Lorentzian interpretation in the three-dimensional space of circles, closely related to the viewpoint used in modern treatments of the Tait–Kneser theorem [1].

The situation changes completely as soon as noncircular ellipses are allowed.

Theorem 1.3 (Elliptic extension)**.** *Let $F = \{P_1, \ldots, P_n\} \subset \mathbb{R}^2$ be any set of $n \geq 1$ distinct points. Then $\mathbb{R}^2 \setminus F$ admits a partition into pairwise disjoint complete curves consisting of exactly $n - 1$ mutually parallel lines together with noncircular ellipses. Consequently, the enlarged problem allowing arbitrary nondegenerate conics and lines has an affirmative answer for every finite F.*

Conversely, if $\mathbb{R}^2 \setminus F$ is partitioned using only ellipses and finitely many lines, then the number of line leaves is necessarily $n - 1$.

The proof of Theorem 1.3 is explicit. Parallel separator lines are chosen so that each complementary strip or half-plane contains exactly one puncture. Each once-punctured cell is then filled by the level conics of a quadratic ratio. For a strip $a < y < b$ containing $p = (u, v)$, one may take

$$\alpha(x - u)^2 + (y - v)^2 = t(y - a)(b - y), \qquad t > 0,$$

with any fixed $0 < \alpha < 1$; every level is a noncircular ellipse contained in the strip, and every point of the punctured strip lies on exactly one level. This construction is developed in Section 7.

Throughout, $B(c, r)$ denotes the open Euclidean disk with center c and radius r.

2 Elementary geometry of disjoint circle leaves

We begin with two elementary facts that will be used repeatedly.

Lemma 2.1 (Nesting of intersecting disks)**.** *Let C_1, C_2 be two distinct Euclidean circles with disjoint circumferences, and let D_1, D_2 be their open disks. If $D_1 \cap D_2 \neq \varnothing$, then one of the closed disks is contained in the other open disk. In particular, if a point p belongs to both D_1 and D_2, then the disks are strictly nested.*

If $D_i = B(c_i, r_i)$ and $\overline{D_1} \subset D_2$, then

$$|c_2 - c_1| + r_1 < r_2.$$

Hence $r_1 < r_2$.

Proof. Two disjoint Euclidean circumferences either bound disjoint disks or one circumference lies in the interior of the other disk. In the latter case compactness of the inner closed disk and positive separation of the two circumferences give strict containment of the closed inner disk in the outer open disk. The displayed inequality is the usual criterion for containment of one closed Euclidean disk in another open disk. □

Lemma 2.2 (Every circle surrounds a missing point)**.** *Let $F \subset \mathbb{R}^2$ be finite, and suppose that $\mathbb{R}^2 \setminus F$ is partitioned into pairwise disjoint complete circles and complete lines. If C is a circle leaf and $D(C)$ is its open disk, then*

$$D(C) \cap F \neq \varnothing.$$

Proof. Assume for contradiction that $D(C_0) \cap F = \varnothing$, where C_0 has center O_0 and radius r_0. Since $O_0 \notin F$, it lies on a leaf. That leaf cannot be a line: every complete line through an interior point of $D(C_0)$ exits the disk and therefore intersects C_0. Hence O_0 lies on another circle leaf C_1.

The circles C_0 and C_1 are disjoint. Since $O_0 \in C_1 \cap D(C_0)$ and $D(C_0)$ is open, the disk $D(C_1)$ meets $D(C_0)$. Lemma 2.1 gives one of the two strict nestings. The alternative $\overline{D(C_0)} \subset D(C_1)$ is impossible, because then the center $O_0 \in D(C_0)$ would belong to $D(C_1)$, whereas $O_0 \in C_1$. Hence

$$\overline{D(C_1)} \subset D(C_0).$$

Let O_1, r_1 be the center and radius of C_1. Since $O_0 \in C_1$, we have $|O_1 - O_0| = r_1$, whereas strict containment gives

$$|O_1 - O_0| + r_1 < r_0.$$

Thus $2r_1 < r_0$.

Iterating, we obtain circle leaves C_k with centers O_k, radii r_k, and

$$\overline{D(C_{k+1})} \subset D(C_k), \qquad r_{k+1} < \frac{1}{2} r_k.$$

The nested compact disks $\overline{D(C_k)}$ have diameters tending to zero, so their intersection is a single point q. Moreover, $q \in D(C_k)$ for every k, because $\overline{D(C_{k+1})} \subset D(C_k)$. In particular $q \in D(C_0)$, so $q \notin F$.

The leaf through q cannot be a line, since a complete line through $q \in D(C_k)$ would intersect C_k. If the leaf through q is a circle A of positive radius, then $q \in A \cap D(C_k)$; since $D(C_k)$ is open, $D(A)$ meets $D(C_k)$. The circle A is disjoint from C_k, so Lemma 2.1 gives one of the two strict nestings. The alternative $\overline{D(C_k)} \subset D(A)$ is impossible because it would place $q \in D(C_k)$ in $D(A)$, while $q \in A = \partial D(A)$. Hence $\overline{D(A)} \subset D(C_k)$. The diameter of A is therefore at most $2r_k$ for every k, impossible because $r_k \to 0$. This contradiction proves the lemma. □

3 The linear skeleton and the bound on the complement

We now study an arbitrary partition of $\mathbb{R}^2 \setminus F$ into complete lines and circles, with F finite.

3.1 The set of line heights

If two distinct complete lines in the plane are disjoint, they are parallel. Hence, whenever line leaves occur, after a rigid motion all of them have the form

$$H_t := \mathbb{R} \times \{t\}.$$

Define

$$S := \{t \in \mathbb{R} : H_t \text{ is a line leaf}\}.$$

Lemma 3.1. *If line leaves occur, then S is closed in $\mathbb{R}$.*

Proof. Let $s_m \in S$ and $s_m \to s$. Suppose $s \notin S$. No term can then equal s. Passing to a subsequence and, if necessary, reflecting the vertical coordinate, we may assume $s_m > s$ for every m. Let

$$q = (x, s) \in H_s \setminus F.$$

Since H_s is not a line leaf, q lies on a circle leaf C_q. If the highest y-coordinate on C_q were strictly larger than s, then for all sufficiently large m the horizontal line H_{s_m} would meet C_q, contrary to disjointness of the partition. Hence the highest point of C_q is exactly q. Thus C_q is tangent to H_s at q and its open disk lies in the half-plane $y < s$.

Different points $q \in H_s \setminus F$ determine different circle leaves, since a circle below H_s has only one highest point. Their open disks are pairwise disjoint. Indeed, if two such disks met, their disjoint circumferences would force one disk to be strictly nested in the other by Lemma 2.1; but the tangency point of the inner circle has height s, whereas every point of the outer open disk has height strictly less than s.

Every nonempty open subset of $\mathbb{R}^2$ contains a point of $\mathbb{Q}^2$, so a pairwise disjoint family of nonempty open subsets of $\mathbb{R}^2$ is at most countable. On the other hand $H_s \setminus F$ is uncountable. This contradiction proves $s \in S$. □

Let I be a connected component of $\mathbb{R} \setminus S$, and put

$$U_I := \mathbb{R} \times I, \qquad F_I := F \cap U_I.$$

Since S is closed, I is an open interval, possibly unbounded. Every point of $U_I \setminus F_I$ lies on a circle leaf. Moreover, every such circle is contained in U_I: if a is a finite endpoint of I, then $a \in S$, so crossing the line H_a would force an intersection with a line leaf.

We shall use the following special case of the main theorem of Moussong and Simányi.

Theorem 3.2 (Moussong–Simányi [2])**.** *Let M be a connected orientable surface without boundary. If M admits a partition into topological circles, then M is homeomorphic either to a torus or to an open annulus $S^1 \times \mathbb{R}$.*

The theorem applies to arbitrary circle decompositions; no foliation hypothesis is required.

Proposition 3.3. *Every connected component I of $\mathbb{R} \setminus S$ contains exactly one point of F, in the sense that*

$$|F_I| = 1.$$

Proof. The surface $U_I \setminus F_I$ is connected, orientable, without boundary, noncompact, and partitioned into circles. By Theorem 3.2, it is homeomorphic to the open annulus. Since $U_I \cong \mathbb{R}^2$, if $k := |F_I|$ then $U_I \setminus F_I$ is homeomorphic to the plane with k punctures. The latter deformation retracts onto a wedge of k circles, so its fundamental group is the free group F_k. The open annulus has fundamental group $\mathbb{Z} = F_1$. Therefore $k = 1$. □

Notice also that the y-coordinate of a point of F cannot belong to S, since the corresponding line would pass through that missing point. Hence the components of $\mathbb{R} \setminus S$ account for all points of F.

3.2 Bounded circular strips are impossible

Lemma 3.4. *No connected component of $\mathbb{R} \setminus S$ is a bounded interval.*

Proof. Assume that $I = (a, b)$ is such a component, and write $w = b - a$. By Proposition 3.3, U_I contains exactly one missing point, say p.

Let C be any circle leaf in U_I, with radius r. Since its entire circumference lies in the strip $a < y < b$, its closed disk also lies in that strip, and hence $2r < w$. By Lemma 2.2, the disk contains a point of F; the only possibility is p. Thus for every $x \in C$,

$$|x - p| < 2r < w.$$

Every point of $U_I \setminus \{p\}$ lies on some circle leaf, so

$$U_I \setminus \{p\} \subset B(p, w),$$

contradicting the horizontal unboundedness of U_I. □

Theorem 3.5 (Finite-complement obstruction)**.** *Let $F \subset \mathbb{R}^2$ be finite. If $\mathbb{R}^2 \setminus F$ can be partitioned into pairwise disjoint complete Euclidean circles and complete affine lines, then*

$$|F| \leq 2.$$

For $|F| = 1$ and $|F| = 2$ such partitions exist.

Proof. First suppose line leaves occur. By Lemma 3.1, S is closed. By Lemma 3.4, every connected component of $\mathbb{R} \setminus S$ is unbounded, so there are at most two such components. Proposition 3.3 says that each contains exactly one point of F, and every point of F lies in one of them. Hence $|F| \leq 2$.

If no line leaf occurs, the whole surface $\mathbb{R}^2 \setminus F$ is decomposed into circles. Theorem 3.2 and the same fundamental-group argument as above give $|F| = 1$.

For one missing point p, the concentric circles $\partial B(p, r)$, $r > 0$, form a partition of $\mathbb{R}^2 \setminus \{p\}$.

For two distinct points p_1, p_2, define

$$f(x) := \frac{|x - p_1|}{|x - p_2|}, \qquad x \in \mathbb{R}^2 \setminus \{p_1, p_2\}.$$

The level sets $f^{-1}(\lambda)$, $\lambda > 0$, partition the domain. For $\lambda = 1$ the level set is the perpendicular bisector of p_1p_2, and for $\lambda \neq 1$ it is an Apollonius circle. Thus the required partition exists. □

Corollary 3.6 (Circle-only case)**.** *For a finite set $F \subset \mathbb{R}^2$, the punctured plane $\mathbb{R}^2 \setminus F$ admits a partition into Euclidean circles if and only if $|F| = 1$.*

Proof. The necessity is the no-line case in Theorem 3.5; the sufficiency is given by concentric circles. □

4 Circle decompositions of a once-punctured plane or half-plane

The remaining geometry occurs in a domain U which is either $\mathbb{R}^2$ or an open half-plane. We first record the local version of the disk-puncture argument needed in this setting; this is logically separate from Lemma 2.2, whose hypothesis concerns a partition of the entire punctured plane by circles and lines.

Lemma 4.1 (Local disk-puncture lemma)**.** *Let U be either $\mathbb{R}^2$ or an open half-plane, let $p \in U$, and suppose that $U \setminus \{p\}$ is partitioned into pairwise disjoint complete Euclidean circles, each contained in U. Then the open disk bounded by every leaf circle contains p.*

Proof. Assume that a leaf circle C_0 bounds an open disk D_0 with $p \notin D_0$. Because U is convex and $C_0 \subset U$, the closed disk $\overline{D_0} = \mathrm{conv}(C_0)$ is contained in U. Hence $D_0 \subset U \setminus \{p\}$.

Let O_0, r_0 be the center and radius of C_0. The point O_0 lies on another leaf circle C_1. Since $O_0 \in C_1 \cap D_0$ and D_0 is open, the disk $D(C_1)$ meets D_0. As C_0, C_1 are disjoint, Lemma 2.1 implies

$$\overline{D(C_1)} \subset D_0.$$

Indeed, the opposite nesting is impossible because it would put the point $O_0 \in C_1$ in the open disk bounded by C_1. If O_1, r_1 are the center and radius of C_1, then $|O_1 - O_0| = r_1$ and strict containment gives

$$|O_1 - O_0| + r_1 < r_0,$$

so $2r_1 < r_0$.

Iterating this construction yields leaf circles C_k with centers O_k and radii r_k such that

$$\overline{D(C_{k+1})} \subset D(C_k), \qquad r_{k+1} < \frac{1}{2} r_k.$$

Their nested closed disks have diameters tending to zero, hence intersect in a single point q. Moreover $q \in D(C_k)$ for every k, and in particular $q \in D_0 \subset U \setminus \{p\}$.

Let A be the leaf circle through q. Since $q \in A \cap D(C_k)$ and A is disjoint from C_k, Lemma 2.1 forces

$$\overline{D(A)} \subset D(C_k)$$

for every k: the reverse containment would place the boundary point $q \in A$ in the open disk $D(A)$. Thus the positive diameter of A is at most $2r_k$ for every k, contradicting $r_k \to 0$. Therefore every leaf disk contains p. □

Fix now $p \in U$, and suppose that $U \setminus \{p\}$ is partitioned into Euclidean circles. By Lemma 4.1, every leaf circle has p in its open disk. Therefore all leaf disks are strictly nested by Lemma 2.1.

Let $R \subset (0, \infty)$ denote the set of radii of the leaf circles. For each $r \in R$, there is at most one leaf of radius r: two distinct leaf disks both containing p would be nested, and strict nesting forces distinct radii. We denote the center of the radius-r leaf by $c(r)$.

4.1 The radius set is all of $(0, \infty)$

Lemma 4.2. *One has*

$$\inf R = 0, \qquad \sup R = \infty.$$

Proof. The intersection of all leaf disks is exactly $\{p\}$. Indeed, p belongs to every leaf disk, while any $x \in U \setminus \{p\}$ lies on its own leaf circumference and hence is not in the corresponding open disk.

Suppose $\rho := \inf R > 0$. We distinguish two cases. If $\rho \in R$, then the radius-ρ leaf disk D_ρ is strictly contained in every leaf disk of larger radius. Hence

$$D_\rho \subseteq \bigcap_{s \in R} D_s,$$

contradicting that the intersection of all leaf disks is exactly $\{p\}$.

Assume therefore that $\rho \notin R$. Choose $r_k \in R$ with $r_k \downarrow \rho$. The nesting inequality gives

$$|c(r_k) - c(r_\ell)| < |r_k - r_\ell|,$$

so the centers form a Cauchy sequence, say $c(r_k) \to c_*$. Passing to the limit in the same inequality yields

$$|c(r_k) - c_*| \leq r_k - \rho.$$

Hence every point of $B(c_*, \rho)$ belongs to every $B(c(r_k), r_k)$. Given any leaf radius $s \in R$, the strict inequality $s > \rho$ and the convergence $r_k \downarrow \rho$ imply $r_k < s$ for all sufficiently large k. Strict nesting then puts $B(c_*, \rho)$ inside the radius-s leaf disk as well. Thus $B(c_*, \rho)$ lies in the intersection of all leaf disks, again contradicting that this intersection is $\{p\}$. Hence $\inf R = 0$.

Suppose instead that $M := \sup R < \infty$. Fix one leaf (c_0, r_0). For any other leaf (c, r), strict nesting gives

$$|c - c_0| < |r - r_0| \leq M + r_0.$$

Thus all centers and all radii are bounded. The union of the leaf circumferences is then bounded, contradicting the fact that $U \setminus \{p\}$ is unbounded. Hence $\sup R = \infty$. □

Lemma 4.3. *The set R is closed relative to $(0, \infty)$.*

Proof. Let $r_k \in R$ and $r_k \to r > 0$. Write $c_k := c(r_k)$. The inequality

$$|c_k - c_\ell| < |r_k - r_\ell|$$

shows that $c_k \to c_*$ for some c_*. Let

$$C_* := \partial B(c_*, r).$$

We claim that C_* is a leaf.

First, p lies strictly inside C_*. Choose $a \in R$ with $a < r$, possible by Lemma 4.2. For all sufficiently large k, $a < r_k$, so

$$|c_k - c(a)| < r_k - a.$$

Passing to the limit gives $|c_* - c(a)| \leq r - a$, and therefore

$$|p - c_*| \leq |p - c(a)| + |c(a) - c_*| < a + (r - a) = r.$$

Also, $C_* \subset \overline{U}$. If U is a half-plane, C_* can meet ∂U in at most one point because its interior contains $p \in U$ and it cannot cross outside $\overline{U}$.

Assume for contradiction that C_* is not a leaf. If some $r_k = r$, then $r \in R$ and there is nothing to prove. Otherwise, after passing to a subsequence, we may suppose either $r_k \uparrow r$ or $r_k \downarrow r$. The set

$$C_* \cap U \setminus \bigcup_k C_{r_k}$$

is uncountable, since each C_{r_k} meets C_* in at most two points. Choose two distinct points q_1, q_2 from this set, and let A_i be the leaf circle through q_i. Each A_i surrounds p.

Suppose first that $r_k \uparrow r$. For fixed k, letting $\ell \to \infty$ in

$$|c_k - c_\ell| < r_\ell - r_k$$

gives $|c_k - c_*| \le r - r_k$, hence $\overline{D_{r_k}} \subset \overline{D_*}$ for $D_* := B(c_*, r)$. By our choice of q_i, the point q_i is not on C_{r_k}; since it lies on C_*, it follows that $q_i \notin \overline{D_{r_k}}$. The disks of A_i and C_{r_k} both contain p, so nesting forces

$$\overline{D_{r_k}} \subset D(A_i)$$

for every k. Passing to the limit yields

$$\overline{D_*} \subset \overline{D(A_i)}.$$

Because q_i lies on both boundaries, A_i is internally tangent to C_* at q_i and lies outside C_*. In particular $A_1 \neq A_2$: one Euclidean circle distinct from C_* cannot be internally tangent to C_* at two different points. If R_i is the radius of A_i and $u_i = (q_i - c_*)/r$, then $R_i > r$ and its center is

$$c_* - (R_i - r)u_i.$$

Since $q_1 \neq q_2$, one has $u_1 \neq u_2$, and consequently the distance between the two centers is strictly larger than

$$|(R_1 - r) - (R_2 - r)| = |R_1 - R_2|.$$

Thus the two disks $D(A_1), D(A_2)$ cannot be nested. But both contain p and their circumferences are disjoint leaf circles, contradicting Lemma 2.1.

If $r_k \downarrow r$, then for fixed k, letting $\ell \to \infty$ in

$$|c_k - c_\ell| < r_k - r_\ell$$

gives $|c_k - c_*| \le r_k - r$, hence $\overline{D_*} \subset \overline{D_{r_k}}$. Since $q_i \notin C_{r_k}$, we have $q_i \in D_{r_k}$. Nesting now forces $\overline{D(A_i)} \subset D_{r_k}$ for every k. Passing to the limit shows that each A_i lies inside C_* and is internally tangent to it at q_i. Again $A_1 \neq A_2$, since a circle distinct from C_* cannot be internally tangent to C_* at two different points. Writing $R_i < r$, the corresponding centers are

$$c_* + (r - R_i)u_i,$$

and again their mutual distance is strictly larger than $|R_1 - R_2|$, contradicting nesting of the two leaf disks. Hence C_* is a leaf and $r \in R$. □

Proposition 4.4. *The set of leaf radii is*

$$R = (0, \infty).$$

Moreover, the center function $c : (0, \infty) \to \mathbb{R}^2$ satisfies

$$|c(s) - c(r)| < s - r \qquad (0 < r < s) \tag{4.1}$$

and

$$\lim_{r \downarrow 0} c(r) = p. \tag{4.2}$$

Proof. By Lemmas 4.2 and 4.3, it remains only to exclude gaps. If (a, b) were a connected component of $(0, \infty) \setminus R$, then $a, b \in R$. The disk D_a is strictly contained in D_b. Choose

$$x \in D_b \setminus \overline{D_a}.$$

The point x lies on some leaf circle of radius s. If $s < a$, strict nesting gives $C_s \subset D_a$, contradicting $x \notin \overline{D_a}$; if $s = a$, then the uniqueness of the radius-a leaf gives $C_s = C_a$, again contradicting $x \notin \overline{D_a}$. If $s = b$, then $x \in D_b = D_s$, contradicting $x \in C_s$; and if $s > b$, strict nesting gives $D_b \subset D_s$, again contradicting $x \in C_s$. Thus $a < s < b$, contrary to the definition of the gap.

Equation (4.1) is precisely the strict disk-containment inequality from Lemma 2.1. Since $p \in B(c(r), r)$ for every r, we have $|c(r) - p| < r$, which immediately gives (4.2). □

4.2 The converse construction

The preceding proposition suggests the correct intrinsic parameter.

Definition 4.5. *Fix $p \in \mathbb{R}^2$. An* admissible center curve anchored at p *is a map*

$$c : (0, \infty) \to \mathbb{R}^2$$

such that

$$\lim_{r \downarrow 0} c(r) = p$$

and

$$|c(s) - c(r)| < s - r \qquad (0 < r < s). \tag{4.3}$$

For such a curve put

$$D_r := B(c(r), r), \qquad C_r := \partial D_r, \qquad K_c := \bigcup_{r>0} D_r.$$

Condition (4.3) is equivalent to

$$\overline{D_r} \subset D_s \qquad (0 < r < s). \tag{4.4}$$

In particular, c is 1-Lipschitz on every compact subinterval of $(0, \infty)$ and extends continuously to $r = 0$ by $c(0) = p$.

Theorem 4.6 (Admissible-curve parametrization)**.** *Let c be an admissible center curve anchored at p. Then the circles*

$$\{C_r : r > 0\}$$

are pairwise disjoint and form a partition of

$$K_c \setminus \{p\}.$$

Conversely, every Euclidean circle decomposition of a once-punctured plane or once-punctured open half-plane arises uniquely in this way, with r equal to the Euclidean radius of the leaf.

Proof. The converse statement is Proposition 4.4, so it remains to prove the construction.

The disks D_r are strictly nested by (4.4), hence K_c is open and convex. We first show that $p \in D_r$ for every $r > 0$. From the 1-Lipschitz extension to 0,

$$|c(r) - p| \leq r.$$

If equality held for some r, then for any $0 < s < r$,

$$r = |c(r) - p| \leq |c(r) - c(s)| + |c(s) - p| < (r - s) + s = r,$$

a contradiction. Thus $|c(r) - p| < r$.

Fix $x \in K_c \setminus \{p\}$ and define

$$\Phi_x(r) := |x - c(r)| - r.$$

The function Φ_x is continuous. If $s > r$, then

$$\Phi_x(s) \leq |x - c(r)| + |c(s) - c(r)| - s < |x - c(r)| - r = \Phi_x(r),$$

so it is strictly decreasing. Moreover,

$$\lim_{r \downarrow 0} \Phi_x(r) = |x - p| > 0.$$

Since $x \in K_c$, some $R > 0$ satisfies $x \in D_R$, so $\Phi_x(R) < 0$. The intermediate value theorem therefore gives a unique $r_x > 0$ such that $\Phi_x(r_x) = 0$, equivalently $x \in C_{r_x}$. Thus the circles partition $K_c \setminus \{p\}$.

Uniqueness of the center curve for a given decomposition follows because each positive radius occurs exactly once by Proposition 4.4. □

Remark 4.7. *The map*

$$\Psi : (0, \infty) \times S^1 \longrightarrow K_c \setminus \{p\}, \qquad \Psi(r, u) = c(r) + ru,$$

is a homeomorphism. Indeed, Theorem 4.6 makes it a continuous bijection. Let $\rho(x)$ denote the unique radius of the leaf through x. To see that ρ is continuous, fix $x \in C_r$ and numbers $0 < a < r < b$. Strict nesting gives

$$x \notin \overline{D_a}, \qquad x \in D_b.$$

Hence some neighborhood V of x is disjoint from $\overline{D_a}$ and contained in D_b. If $y \in V$ lies on $C_{\rho(y)}$, then $\rho(y) \leq a$ would imply $y \in \overline{D_a}$, while $\rho(y) \geq b$ would imply $y \in D_{\rho(y)}$, contradicting $y \in C_{\rho(y)}$. Thus

$$a < \rho(y) < b$$

throughout V, proving continuity of ρ. The inverse map is therefore

$$x \longmapsto \left(\rho(x), \frac{x - c(\rho(x))}{\rho(x)}\right),$$

which is continuous. Thus, although arbitrary circle decompositions of surfaces need not be foliations in general [2], these Euclidean decompositions have a global topological product parametrization.

5 The domain dichotomy

A priori, the increasing union K_c of disks could be an arbitrary unbounded convex domain. The strict contraction condition rules this out completely.

Theorem 5.1 (Plane–half-plane dichotomy)**.** *For every admissible center curve c, the domain K_c is either the entire plane or an open half-plane.*

Proof. Assume $K_c \neq \mathbb{R}^2$. Since K_c is a nonempty open convex set, choose $q \in \partial K_c$ and a supporting line to K_c at q. Let ν be the unit normal pointing into the supporting half-plane, so

$$K_c \subset H := \{x \in \mathbb{R}^2 : \nu \cdot (x - q) > 0\}.$$

For every $r > 0$, the compact circle C_r is contained in K_c because $C_r \subset D_s$ for every $s > r$. Hence

$$\varepsilon(r) := \nu \cdot (c(r) - q) - r > 0.$$

If $s > r$, then

$$\varepsilon(s) - \varepsilon(r) = \nu \cdot (c(s) - c(r)) - (s - r) \leq |c(s) - c(r)| - (s - r) < 0.$$

Thus ε is strictly decreasing and has a limit $\varepsilon_\infty \geq 0$. If $\varepsilon_\infty > 0$, then every point of every D_r has signed distance $> \varepsilon_\infty$ from the supporting line, contradicting $q \in \partial K_c$. Therefore

$$\varepsilon(r) \downarrow 0. \tag{5.1}$$

Choose orthonormal coordinates with $q = 0$ and $\nu = (0, 1)$, and write

$$c(r) = (a(r), r + \varepsilon(r)).$$

For $r > R$, condition (4.3) gives

$$(a(r) - a(R))^2 + \big((r - R) - (\varepsilon(R) - \varepsilon(r))\big)^2 < (r - R)^2.$$

Consequently,

$$|a(r) - a(R)|^2 < 2(r - R)(\varepsilon(R) - \varepsilon(r)) \leq 2r\varepsilon(R).$$

For fixed R this yields

$$\limsup_{r\to\infty} \frac{|a(r)|}{\sqrt{r}} \leq \sqrt{2\varepsilon(R)}.$$

Letting $R \to \infty$ and using (5.1), we obtain

$$a(r) = o(\sqrt{r}). \tag{5.2}$$

Now take any $x = (u, v) \in H$, so $v > 0$. A direct expansion gives

$$|x - c(r)|^2 - r^2 = (u - a(r))^2 - 2r(v - \varepsilon(r)) + (v - \varepsilon(r))^2.$$

Dividing by r and using (5.1) and (5.2), the right-hand side tends to $-2v < 0$. Hence $x \in D_r$ for all sufficiently large r. Thus $H \subset K_c$. The reverse inclusion was built into the supporting half-plane, so $K_c = H$. □

The two cases can be detected by a single monotone scalar quantity.

Theorem 5.2 (Escape-margin criterion)**.** *Let c be admissible and define*

$$d(r) := r - |c(r) - p|.$$

Then $d(r) > 0$ and d is strictly increasing. Its limit

$$\delta := \lim_{r\to\infty} d(r)$$

exists in $(0, \infty]$. Moreover,

$$K_c = \mathbb{R}^2 \quad \Longleftrightarrow \quad \delta = \infty.$$

If $\delta < \infty$, then K_c is an open half-plane and

$$\operatorname{dist}(p, \partial K_c) = \delta.$$

Proof. We already know $p \in D_r$, so $d(r) > 0$. If $0 < r < s$, then

$$|c(s) - p| \leq |c(s) - c(r)| + |c(r) - p| < (s - r) + |c(r) - p|,$$

which is equivalent to $d(s) > d(r)$.

Since $B(p, d(r)) \subset D_r$, if $d(r) \to \infty$ then $K_c = \mathbb{R}^2$. Conversely, suppose $K_c = \mathbb{R}^2$. For any $R > 0$, the compact disk $\overline{B(p, R)}$ is covered by the nested open family $\{D_r : r > 0\}$. A finite subcover and nesting give some r with

$$\overline{B(p, R)} \subset D_r.$$

The distance from p to $\mathbb{R}^2 \setminus D_r$ is exactly $d(r)$, so $d(r) > R$. Since R is arbitrary, $d(r) \to \infty$.

Finally suppose $\delta < \infty$. By Theorem 5.1, K_c is an open half-plane. Let

$$D := \operatorname{dist}(p, \partial K_c).$$

The inclusion $B(p, d(r)) \subset K_c$ gives $d(r) \leq D$, hence $\delta \leq D$. Conversely, if $0 < R < D$, then $\overline{B(p, R)} \subset K_c$. The same compactness-and-nesting argument gives some r with $\overline{B(p, R)} \subset D_r$, hence $d(r) > R$. Therefore $\delta \geq R$ for every $R < D$, and so $\delta \geq D$. Thus $\delta = D$. □

Proposition 5.3 (Asymptotics in the half-plane case)**.** *Assume K_c is a half-plane. Let ν be its inward unit normal, let*

$$\delta = \operatorname{dist}(p, \partial K_c), \qquad q = p - \delta\nu \in \partial K_c,$$

and let τ be a unit tangent vector to ∂K_c. Then

$$\frac{c(r) - p}{|c(r) - p|} \longrightarrow \nu.$$

Moreover, one can write

$$c(r) = q + a(r)\tau + (r + \varepsilon(r))\nu,$$

where

$$\varepsilon(r) > 0, \qquad \varepsilon(r) \to 0, \qquad a(r) = o(\sqrt{r}).$$

Proof. Since every circle C_r is compactly contained in the half-plane,

$$\nu \cdot (c(r) - p) + \delta - r > 0,$$

or equivalently

$$r - \nu \cdot (c(r) - p) < \delta.$$

On the other hand,

$$d(r) = r - |c(r) - p| \leq r - \nu \cdot (c(r) - p).$$

Theorem 5.2 gives $d(r) \to \delta$, and hence

$$r - \nu \cdot (c(r) - p) \to \delta.$$

Subtracting the two quantities yields

$$|c(r) - p| - \nu \cdot (c(r) - p) \to 0.$$

Since $|c(r) - p| = r - d(r) \to \infty$, division by $|c(r) - p|$ shows

$$\nu \cdot \frac{c(r) - p}{|c(r) - p|} \to 1,$$

which implies the stated directional convergence.

The representation follows by resolving $c(r) - q$ into the orthonormal basis (τ, ν). The positivity and convergence of ε are exactly the supporting-line calculation in the proof of Theorem 5.1, and the estimate $a(r) = o(\sqrt{r})$ is equation (5.2). □

Corollary 5.4 (Criterion for a prescribed half-plane)**.** *Let*

$$H = \{x \in \mathbb{R}^2 : \nu \cdot (x - q) > 0\},$$

where ν is a unit vector, and let $p \in H$. For an admissible center curve c anchored at p, define

$$\varepsilon_H(r) := \nu \cdot (c(r) - q) - r.$$

Then $K_c = H$ if and only if

$$\varepsilon_H(r) > 0 \quad \textit{for every } r > 0, \qquad \varepsilon_H(r) \longrightarrow 0 \quad (r \to \infty).$$

Proof. If $K_c = H$, every compact circle C_r lies in H, so $\varepsilon_H(r) > 0$; the supporting-line argument in the proof of Theorem 5.1 gives $\varepsilon_H(r) \to 0$.

Conversely, assume the two displayed conditions. The inequality $\varepsilon_H(r) > 0$ says precisely that $\overline{D_r} \subset H$ for every r, hence $K_c \subset H$. Therefore $K_c \neq \mathbb{R}^2$, so Theorem 5.1 shows that K_c is an open half-plane. Any open half-plane contained in H must have boundary parallel to ∂H and hence is of the form

$$\{x : \nu \cdot (x - q) > \alpha\}$$

for some $\alpha \geq 0$. On the other hand,

$$\inf_{x \in D_r} \nu \cdot (x - q) = \varepsilon_H(r),$$

and therefore

$$\inf_{x \in K_c} \nu \cdot (x - q) = \inf_{r > 0} \varepsilon_H(r) = 0.$$

Thus $\alpha = 0$, and $K_c = H$. □

6 Complete classification for finite complement

We can now combine the linear skeleton with the admissible-curve theorem.

Theorem 6.1 (Global classification)**.** *Let $F \subset \mathbb{R}^2$ be finite, and suppose $\mathbb{R}^2 \setminus F$ is partitioned into pairwise disjoint complete Euclidean circles and complete affine lines. Up to a Euclidean isometry, exactly one of the following four types occurs.*

(I) ***No punctures.*** *$F = \varnothing$, there are no circle leaves, and the partition is*

$$\{\mathbb{R} \times \{t\} : t \in \mathbb{R}\}.$$

(II) ***One puncture, pure circle type.*** *$F = \{p\}$ and there are no line leaves. The partition is*

$$\{\partial B(c(r), r) : r > 0\},$$

where c is an admissible center curve anchored at p satisfying

$$r - |c(r) - p| \longrightarrow \infty.$$

Conversely every such curve gives such a partition.

(III) ***One puncture, half-plane type.*** *For some $p = (p_1, p_2)$ with $p_2 > 0$,*

$$F = \{p\}.$$

The line leaves are exactly

$$\{\mathbb{R} \times \{t\} : t \leq 0\},$$

and the upper half-plane $\{y > 0\} \setminus \{p\}$ is partitioned by the circles of an admissible center curve $c = (c_1, c_2)$ anchored at p which generates $\{y > 0\}$. By Corollary 5.4, the latter condition is equivalent to

$$c_2(r) - r > 0 \quad (r > 0), \qquad c_2(r) - r \longrightarrow 0.$$

In particular, Theorem 5.2 gives

$$r - |c(r) - p| \longrightarrow p_2 = \operatorname{dist}(p, \{y = 0\}).$$

Conversely every admissible curve satisfying the two conditions from Corollary 5.4, together with the indicated lines, gives a partition.

(IV) ***Two punctures.*** *For some $w \geq 0$,*

$$F = \{p_-, p_+\}, \qquad p_- \in \{y < 0\}, \quad p_+ \in \{y > w\}.$$

The line leaves are exactly

$$\{\mathbb{R} \times \{t\} : 0 \leq t \leq w\}.$$

The lower half-plane $\{y < 0\} \setminus \{p_-\}$ and the upper half-plane $\{y > w\} \setminus \{p_+\}$ are independently partitioned by admissible center curves generating those two half-planes. Equivalently, Corollary 5.4 applies to each side with the appropriate inward unit normal. Conversely every such pair of curves, together with the indicated line leaves, gives a partition.

No other finite-complement partition exists.

Proof. If $F = \varnothing$, Lemma 2.2 rules out circle leaves. Any two line leaves are parallel, and covering the whole plane forces every parallel translate to occur. This is Type (I).

Assume now $|F| \geq 1$. By Theorem 3.5, $|F| \leq 2$.

If $|F| = 1$ and there are no lines, Theorem 4.6 and Theorem 5.2 give Type (II).

Suppose line leaves occur. Use horizontal coordinates and the closed height set S. Proposition 3.3 says that every component of $\mathbb{R} \setminus S$ contains exactly one puncture, while Lemma 3.4 says that every such component is unbounded.

If $|F| = 1$, the complement $\mathbb{R} \setminus S$ therefore consists of exactly one unbounded interval. Since S is nonempty and closed, after reflection and vertical translation we have

$$S = (-\infty, 0].$$

The unique circular component is the open upper half-plane, and Theorems 4.6 and 5.2 give precisely Type (III).

If $|F| = 2$, the complement $\mathbb{R} \setminus S$ consists of exactly two unbounded intervals. Thus for some $a \le b$,

$$S = [a, b].$$

After vertical translation write $a = 0$ and $b = w \ge 0$. Each of the two complementary half-planes contains exactly one puncture, and Theorems 4.6 and 5.2 independently classify its circle leaves. This is Type (IV).

The converse assertions follow directly from Theorem 4.6: the circular families partition their respective open regions, the horizontal lines partition the remaining closed line region, and the two parts are disjoint. □

Remark 6.2 (The Apollonius family)**.** *In Type (IV), the case $w = 0$ has a single line leaf separating the two circular half-planes. The standard construction obtained from the level sets*

$$\frac{|x - p_-|}{|x - p_+|} = \lambda$$

is one highly symmetric example: the level $\lambda = 1$ is the separating line and the remaining levels are Apollonius circles. Theorem 6.1 shows that this example is far from unique.

7 Elliptic extension: arbitrary finite complements

The circle-and-line obstruction $|F| \le 2$ is destroyed as soon as noncircular ellipses are admitted. The key point is that a bounded-width strip can be exhausted by ellipses whose minor axes remain bounded while their major axes diverge. We first give the local construction.

7.1 A once-punctured strip

Proposition 7.1 (Elliptic strip decomposition)**.** *Let*

$$U = \{(x, y) \in \mathbb{R}^2 : a < y < b\}, \qquad a < b,$$

and let $p = (u, v) \in U$. Fix $0 < \alpha < 1$, and set

$$A(x, y) = \alpha(x - u)^2 + (y - v)^2, \qquad B(x, y) = (y - a)(b - y).$$

For $t > 0$, define

$$E_t := \{(x, y) \in \mathbb{R}^2 : A(x, y) = tB(x, y)\}.$$

Then every E_t is a nondegenerate noncircular ellipse contained in U, the ellipses E_t are pairwise disjoint, and

$$U \setminus \{p\} = \bigsqcup_{t>0} E_t.$$

Proof. For $t > 0$ the defining polynomial is

$$A - tB = \alpha(x - u)^2 + (y - v)^2 - t(y - a)(b - y).$$

Its quadratic part is

$$\alpha x^2 + (1 + t)y^2,$$

which is positive definite. At the point p one has

$$(A - tB)(p) = -t(v - a)(b - v) < 0.$$

A real quadratic polynomial with positive-definite quadratic part has compact ellipsoidal zero set whenever its minimum is negative; hence E_t is a nondegenerate ellipse. It is not a circle, because the eigenvalues α and $1 + t$ of its quadratic part are distinct.

If $(x, y) \notin U$, then $B(x, y) \leq 0$ while $A(x, y) \geq 0$. Thus $A = tB$ would force $A = B = 0$. But $A = 0$ only at $p \in U$, so this is impossible. Hence $E_t \subset U$.

Now let $q \in U \setminus \{p\}$. Then $A(q) > 0$ and $B(q) > 0$, so the unique parameter

$$t(q) = \frac{A(q)}{B(q)} > 0$$

satisfies $q \in E_{t(q)}$. Conversely, if a point belonged to both E_{t_1} and E_{t_2}, then $B > 0$ at that point and therefore $t_1 = A/B = t_2$. Thus the family is pairwise disjoint and covers exactly $U \setminus \{p\}$. □

7.2 A once-punctured half-plane

Proposition 7.2 (Elliptic half-plane decomposition)**.** *Let*

$$U = \{(x, y) \in \mathbb{R}^2 : y > a\}, \qquad p = (u, v) \in U,$$

and fix $0 < \alpha < 1$. *Put*

$$A(x, y) = \alpha(x - u)^2 + (y - v)^2, \qquad B(x, y) = y - a.$$

Then the level sets

$$E_t := \{A = tB\}, \qquad t > 0,$$

are pairwise disjoint nondegenerate noncircular ellipses contained in U *and form a partition of* $U \setminus \{p\}$.

Proof. The quadratic part of $A - tB$ is $\alpha x^2 + y^2$, hence positive definite, while

$$(A - tB)(p) = -t(v - a) < 0.$$

Thus each zero set is a nondegenerate ellipse; it is noncircular because $\alpha \neq 1$. If $y \leq a$, then $B \leq 0$ whereas $A \geq 0$, so the equation $A = tB$ has no solution there. Finally every point $q \in U \setminus \{p\}$ lies on the unique level with $t = A(q)/B(q) > 0$. □

The lower-half-plane version is obtained by replacing $B = y - a$ with $B = b - y$. For the whole plane with one puncture $p = (u, v)$, the simpler family

$$\alpha(x - u)^2 + (y - v)^2 = t, \qquad t > 0,$$

provides a partition by similar noncircular ellipses.

7.3 The global construction

Theorem 7.3 (Arbitrary finite complements by ellipses and lines)**.** *Let*

$$F = \{P_1, \dots, P_n\} \subset \mathbb{R}^2, \qquad n \geq 1,$$

be any set of distinct points. Then $\mathbb{R}^2 \setminus F$ admits a partition into pairwise disjoint complete curves consisting of exactly $n-1$ mutually parallel lines and noncircular ellipses.

Proof. If $n = 1$, write $P_1 = (u, v)$ and fix any $0 < \alpha < 1$. The ellipses

$$\alpha(x-u)^2 + (y-v)^2 = t, \qquad t > 0,$$

partition $\mathbb{R}^2 \setminus \{P_1\}$, so the assertion holds with no line leaves. Assume henceforth that $n \geq 2$.

Choose a unit vector e such that the linear functional

$$\ell(x) = e \cdot x$$

takes pairwise distinct values on F. Such a direction exists because, for each pair $i \neq j$, the forbidden condition

$$e \cdot (P_i - P_j) = 0$$

excludes only two antipodal unit directions, and there are finitely many pairs.

Relabel the points so that

$$\ell(P_1) < \ell(P_2) < \cdots < \ell(P_n).$$

For $1 \leq i \leq n-1$, choose

$$s_i \in \big(\ell(P_i), \ell(P_{i+1})\big)$$

and let

$$L_i = \{x \in \mathbb{R}^2 : \ell(x) = s_i\}.$$

Since $n \geq 2$, these $n-1$ lines are mutually parallel and divide the plane into n open cells: two half-planes and, when $n \geq 3$, $n-2$ finite-width strips. By construction, each cell contains exactly one point of F.

After a Euclidean isometry, each cell is of the form treated in Proposition 7.1 or Proposition 7.2. Fill every punctured cell by the corresponding family of noncircular ellipses, and add the separator lines $L_1, \dots, L_{n-1}$. Ellipses belonging to different cells are disjoint because the cells are disjoint, no ellipse meets a separator line, and the separator lines themselves are disjoint. The union of all leaves is exactly $\mathbb{R}^2 \setminus F$. □

Since every ellipse is a nondegenerate conic, Theorem 7.3 immediately gives the existence statement for the enlarged problem in which all nondegenerate ellipses, parabolas, hyperbolas, circles, and lines are allowed.

The number of line leaves in the preceding construction is not merely convenient; within the ellipse-and-finite-lines class it is forced.

Theorem 7.4 (Optimality of the line count)**.** *Let $F \subset \mathbb{R}^2$ have $n \geq 1$ points. Suppose $\mathbb{R}^2 \setminus F$ is partitioned using only complete ellipses and finitely many complete lines. If there are m line leaves, then*

$$m = n - 1.$$

In particular, Theorem 7.3 uses the minimum possible number of line leaves in this class.

Proof. Any two line leaves are disjoint, hence parallel. Thus m line leaves divide the plane into exactly $m+1$ open cells, each homeomorphic to $\mathbb{R}^2$. No puncture lies on a line leaf, so the punctures are distributed among these cells. Inside each cell there are no line leaves; the punctured cell is therefore partitioned entirely by ellipses, hence by topological circles.

Let one cell contain k punctures. The cell is homeomorphic to $\mathbb{R}^2$, so after removing those k points its fundamental group is the free group F_k. By Theorem 3.2, a connected orientable boundaryless surface decomposed into topological circles and homeomorphic to a noncompact surface must be an open annulus, whose fundamental group is $\mathbb{Z}$. Hence $F_k \cong \mathbb{Z}$, so $k = 1$. Therefore every one of the $m + 1$ cells contains exactly one puncture, and

$$n = m + 1.$$

Equivalently, $m = n - 1$. □

Remark 7.5 (Quadratic-ratio interpretation)**.** *The local constructions above are level sets of a rational quadratic function*

$$\Phi = \frac{A}{B}.$$

In the strip case, A is a positive-definite quadratic form vanishing only at the puncture and $B = (y - a)(b - y)$ vanishes on the two boundary lines. Thus the pencil $A - tB = 0$ interpolates between a point degeneration at $t = 0$ and the pair of boundary lines at the projective endpoint $t = \infty$. The essential new geometric freedom, absent for circles, is that the eccentricity may tend to 1 as $t \to \infty$, allowing the ellipses to remain inside a bounded-width strip while extending arbitrarily far in the longitudinal direction.

8 Moduli and the Lorentzian circle space

The classification is functional rather than finite-dimensional. For example, fix a unit vector e, a point p, and $0 < a < 1$. The center curve

$$c(r) = p + a(1 - e^{-r})e$$

is admissible, because it is differentiable with $|c'(r)| = ae^{-r} < 1$, and

$$r - |c(r) - p| \longrightarrow \infty.$$

It therefore gives a nonconcentric circle decomposition of $\mathbb{R}^2 \setminus \{p\}$.

Similarly, let $q \in \mathbb{R}^2$, let ν be a unit vector, and let $\delta > 0$. Put

$$p = q + \delta\nu, \qquad c(r) = q + \big(r + \delta e^{-r/\delta}\big)\nu.$$

Then $c(r) \to p$ as $r \downarrow 0$ and

$$|c'(r)| = 1 - e^{-r/\delta} < 1.$$

Moreover

$$r - |c(r) - p| = \delta(1 - e^{-r/\delta}) \longrightarrow \delta.$$

Hence this family partitions the half-plane

$$\{x : \nu \cdot (x - q) > 0\}$$

minus the point p.

There is also a useful Lorentzian reformulation. Encode a circle with center $c = (a, b)$ and radius r by the point $(a, b, r) \in \mathbb{R}^{2,1}$, equipped with the quadratic form

$$Q(\Delta c, \Delta r) = (\Delta r)^2 - |\Delta c|^2.$$

For $s > r$, the strict nesting condition is exactly

$$Q(c(s) - c(r), s - r) > 0$$

with positive time orientation. Thus the graph

$$\Gamma(r) = (c(r), r)$$

has every forward chord timelike. Lorentzian geometry of the space of circles is also the mechanism behind a proof of the Tait–Kneser theorem in [1]; here it appears not for osculating circles of a prescribed plane curve, but as the intrinsic geometry of all nested circle decompositions considered above.

9 Concluding remarks

The circle-and-line problem is rigid at the level of topology and domain geometry but flexible at the level of individual circular families. The topology forces at most two missing points and confines all straight leaves to a parallel line skeleton. Euclidean nesting then reduces every circular component to a once-punctured plane or half-plane, while the admissible center curve retains an infinite-dimensional metric freedom.

The elliptic extension shows that this obstruction is not a generic feature of conic geometry. Once noncircular ellipses are admitted, every finite puncture set becomes realizable, and an explicit quadratic-ratio construction produces a partition using exactly $n - 1$ parallel lines. In the ellipse-and-finitely-many-lines class this line count is forced. The contrast isolates the decisive geometric difference: circles have a single isotropic scale, whereas ellipses can keep a bounded transverse width while their longitudinal scale diverges.

Natural next problems include classifying finite-complement partitions when lines are forbidden but general nondegenerate conics are allowed; determining what can be done using only parabolas and/or hyperbolas together with lines; understanding the separator structure created by noncompact conics; and classifying those conic partitions that arise from quadratic pencils or, more generally, rational quadratic first integrals. For the circular subproblem, one may also study admissible center curves modulo similarities or Möbius transformations, impose smooth or analytic regularity, or seek higher-dimensional analogues involving hyperspheres and hyperplanes.

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